

Data-free Neural Representation Compression with Riemannian Neural Dynamics

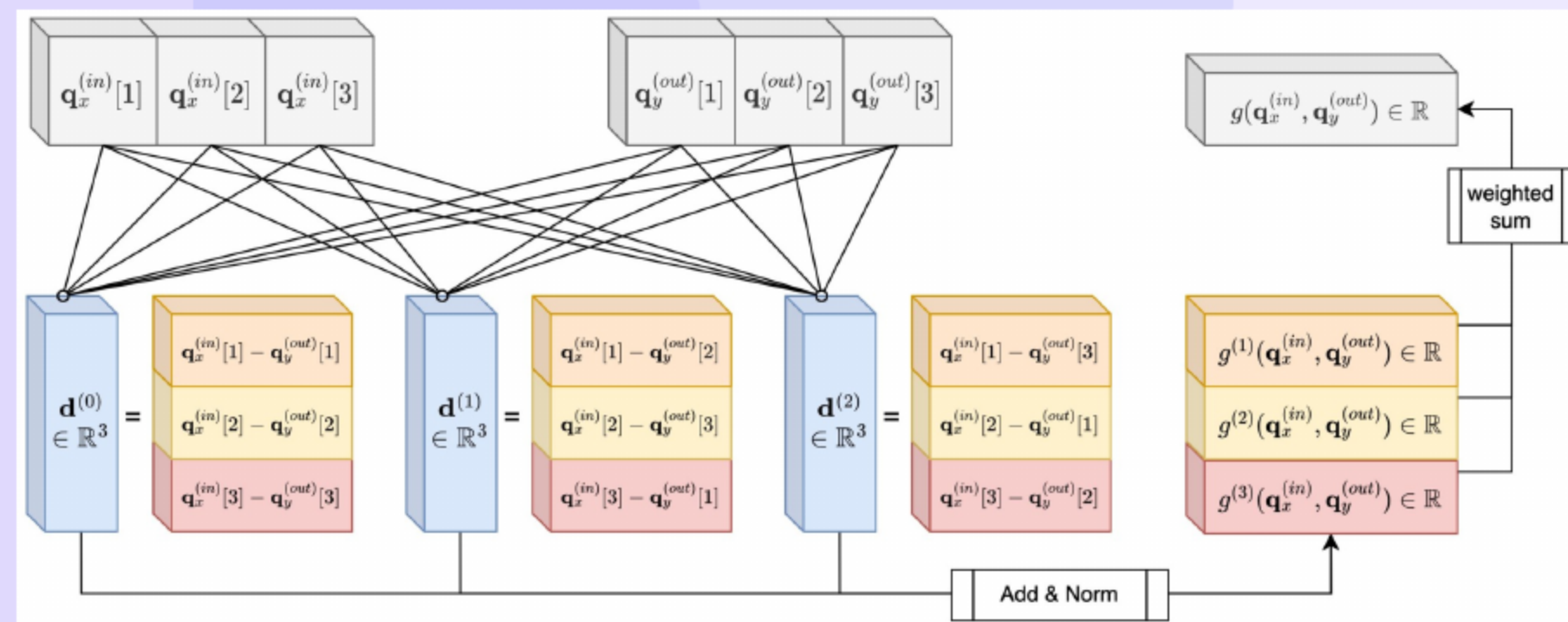
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01 Paper Title and Authors

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- Affiliations as listed
 - Labels one through five per paper header
- Paper title
 - Data-free Neural Representation Compression with \Riemannian Neural Dynamics (\RieM)
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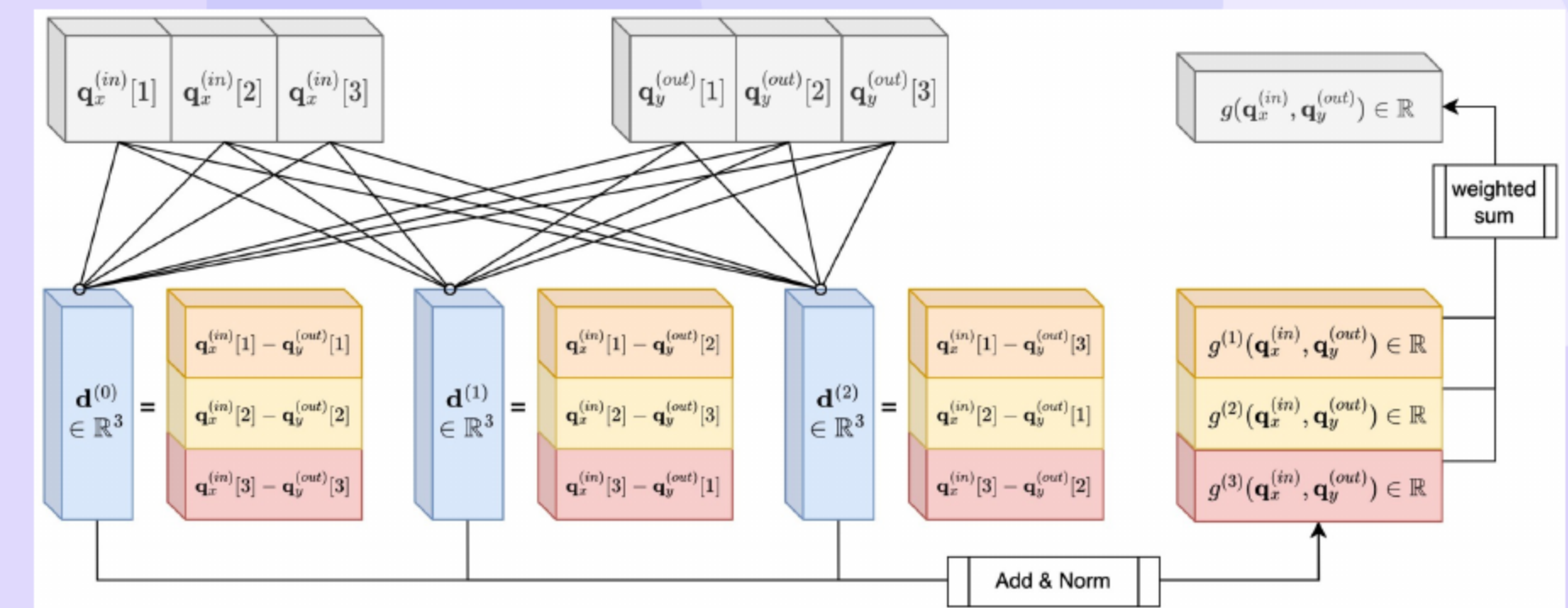
RieM: Reinterpreting Neural Computation via Riemannian Neuronal State Space



- Core idea reframes weights as metric-driven interactions between neuron states with **compact parameters**.
- Highlights include **higher nonlinearity**, **parameter efficiency**, and no data fine-tuning.
- Results show LeNet-5 is 8.5 times smaller with **small accuracy gain** on MNIST.
- Results show Swin-T is four times smaller with **minor accuracy drop** on ImageNet.
- Results show DETR is 4.8 times smaller with **slight AP gain** on COCO.
- Why now: tight on-device budgets make **interaction-preserving** compression crucial.

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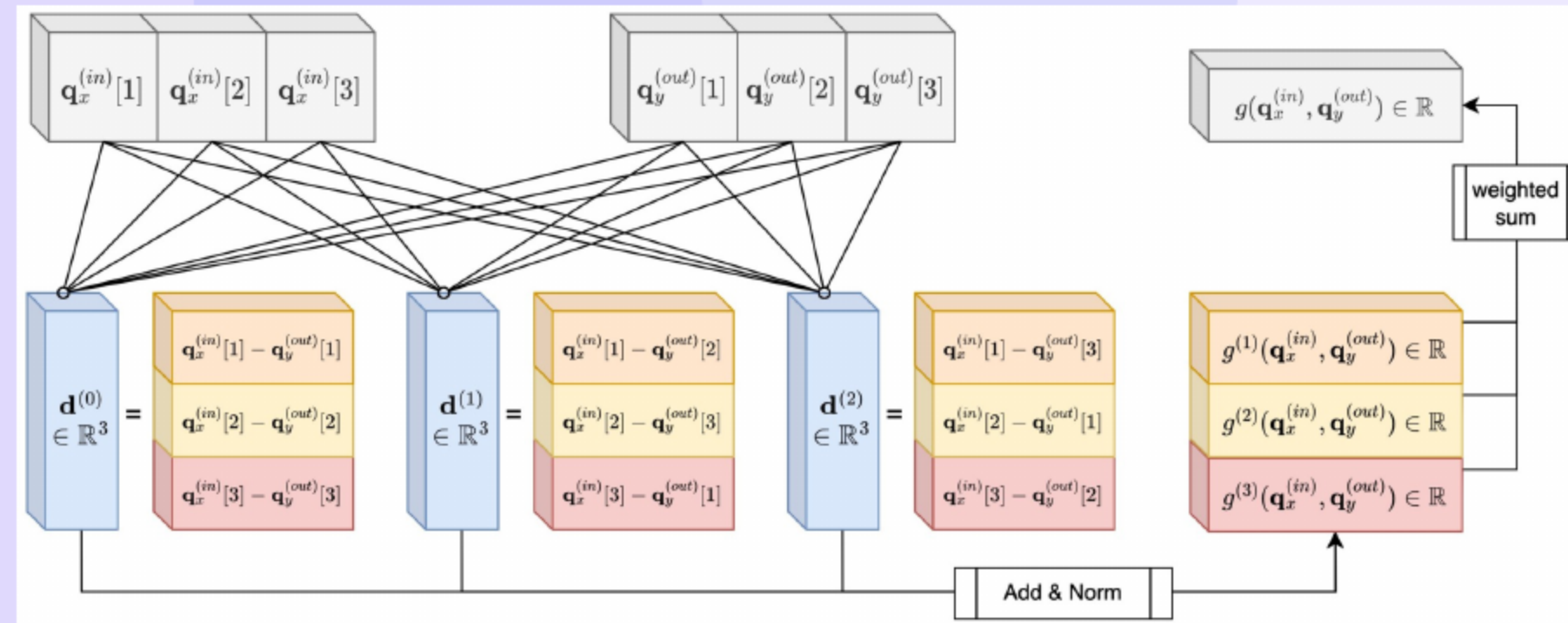
Introduction: From Weight-Based Models to Riemannian Neuron Dynamics (RieM)



- Motivation emphasizes that weight-space compression limits **cross-neuron interactions** under tight budgets.
- RieM improves expressivity by modeling dense couplings with **learned metrics** over neuron states.
- Positioning differs from pruning and quantization by reparameterizing **interaction structure**.
- Practical components include displacement modeling, shared structure use, and **dynamic merging**.
- Empirical scope covers CNNs, Transformers, and detection with **better trade-offs**.
- Why weight-space can fail under tight budgets: low-rank and quant-prune bottleneck **nonlinearity** and **cross-neuron coupling**.

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RieM Mechanics: Metric-Based Neuron Dynamics and Sub-metric Aggregation



- Mapping computes each weight from a metric applied to **neuron embeddings** and metric parameters.
- Option Two uses displacement vectors, projections, and a nonlinear merge for **stronger fitting**.
- Sub-metrics aggregate multiple metric components with **learned weights** for added capacity.
- Expressive interactions enable **lower fitting error** than feed-forward or low-rank forms.

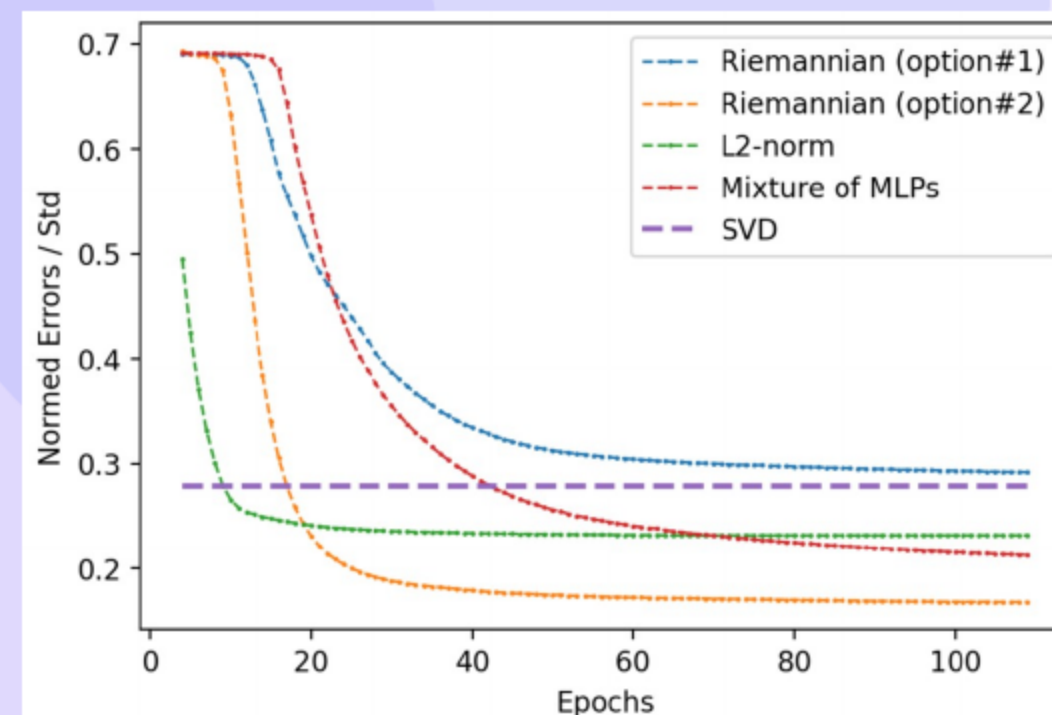
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Most \textbf{Surprising} Result: Metric Interactions Beat Data-Free Baselines.

- Swin-T (8/8): RieM **80.85\%** vs PSAQ-ViT V2 80.21\% \u2192 **+0.64** top-1.
- DETR (R50, 8-bit): **-0.4 AP** vs Quant-DETR **-2.2**.
- LeNet-5: **8.5\%** smaller with **+0.18\%** on MNIST (data-free).

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Neural Compression in Metric Space: Method and SCC-Guided Multi-Matrix Strategy



- Objective minimizes reconstruction error with compact neuron states and **few metric parameters**.
- SCC strategy concatenates matrices using correlation counts with **greedy pairing** decisions.
- ImageNet results surpass DFQ and Lp-norm with **strong accuracy** at similar sizes.
- Shared structure across layers improves **compression fidelity** under strict budgets.
- Compute & training: **dyMerge** with a learned threshold; tune $\mathbf{Dq}, \mathbf{D}\mu$; **no real data**.

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Results: ViT + ImageNet CNNs

- **Swin-T** (4/8, 14.5 MB, 144G BOPs): **RieM76.30\%** vs **PSAQ-ViT V2** 76.28\%; (8/8): **80.85\%** vs 80.21\%.
- **Swin-S** (4/8, 25 MB, 278G): **RieM79.84\%** vs 78.86\%; (8/8): **82.96\%** vs 82.13\%.
- **ImageNet-1k CNNs**: at equal size, **ResNet-50** is **+2.2\%** over **SQuant** and **+1.2\%** over **UDFC**.
- At **12.28 MB**: **RieM73.26\%** vs **Lp-norm** 72.96\% (similar compute).

- **DETR** (R50, 8-bit): AP drop **-0.4** at 43.6 MB; better than **Quant-DETR-2.2** and **SVD-DETR-11.5**.
- **Equal sizes**: **RieM** beats **T-DETR** on 5/6 AP metrics (APS exception).
- **Pruning**: **ResNet-101** at 30\% prune is **+34.8\%** over **NM** and **+7.5\%** over **UDFC**.
- **Ablations**: **Option Two** yields **lower** fitting error; **SCC** and **dyMerge** improve fidelity/speed.

Thank You

Questions & Discussion